ARYAN MEENA

Data Scientist and Machine Learning Engineer | Python | SQL | PyTorch | AWS
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EXPERIENCE

Data Scientist (Research Assistantship) | Boston University Sep 2025 – May 2026

- Forecast 5-day currency volatility out of sample at R^2 0.25 vs a 0.05 noise ceiling across 8 CME futures time series, testing 6 statistical models and machine learning methods under purged walk-forward validation.
- Caught 5 feature engineering bugs before fitting, one of which would have leaked 615 zeros into the targets.
- Traced a 15-year trading signal to biased opening prices in a free data feed with statistical analysis against exchange data (Sharpe -1.36 vs +0.25), and the research paper is now **published in Risks (MDPI)**.
- Wrote the SQL and Python behind a live 12-table dashboard where every chart is a query anyone can rerun.
- Rewrote the Python backtest engine in NumPy to run 1,312 configurations 17x faster with identical results.

Graduate Teaching Assistant | Boston University Jan 2026 – May 2026

- Coached 60+ graduate students training as data scientists in Advanced Machine Learning and Data Science with Python, explaining statistics and model results in plain language and working through their code bugs.

Machine Learning Research Intern | ISRO Space Applications Centre Jun 2022 – Jul 2022

- Trained U-Net and DeepLabV3+ deep learning models in PyTorch to segment roads in 6,723 aerial images.

PROJECTS

Hospital Readmission Risk | Python, LightGBM, PyTorch, Tableau Care Dashboard

- Beat a published benchmark (**AUC 0.689 vs 0.664**) with classification models on 101,766 diabetes encounters, picking LightGBM over XGBoost and logistic regression under patient-grouped cross-validation.
- Shipped LightGBM after model development showed two PyTorch Transformer models cost up to 65x more to train.
- Ranked patients by risk so the top 5% held **3.0x** the readmissions of a random 5% in a Tableau data visualization.

JudgeGuard: LLM Answer Checking | Python, FastAPI, Docker Live App

- Evaluated a local large language model (Qwen3-4B) as a natural language processing (NLP) answer grader: it ranked the source-backed answer first on 96 of 100 pairs (95% CI 92-99%), with every raw response published.
- Owned model deployment for this generative AI application: a FastAPI service in a Docker image and a live Hugging Face app, with 150 tests under Git and CI/CD and a 150 ms p99 latency budget.

Service Operations Analytics | Python, SQL, AWS, GitHub Actions Live Dashboard

- Automated a daily AWS data pipeline (130K+ records) that publishes only after **17 data quality checks** pass.
- Picked the forecasting model by A/B testing: a challenger must cut error 2%+ in 2 of 3 windows to win.
- Set up daily model monitoring of forecast quality for 5 NYC boroughs with automated alarms.

EDUCATION

Master of Science (MS), Applied Data Analytics, Boston University (GPA 3.75/4.0) May 2026

- Led a six-person team to a BU MedAI Hackathon podium among 40+ teams, and first-authored 5 publications.
- Coursework: Foundations of Machine Learning, Advanced Machine Learning, Big Data Analytics, Data Mining.

Bachelor of Technology, Electronics and Communication Engineering, IIST, India May 2023

- Coursework: Statistics and Probability, Computer Vision, Machine Learning for Signal Processing.

SKILLS

Programming: Python (pandas, NumPy, scikit-learn), SQL, PyTorch, Spark (PySpark), Git, Docker, FastAPI, R
Machine learning: deep learning, gradient boosting (XGBoost, LightGBM), NLP, LLM evaluation, time series, statistics
MLOps and data: model deployment, model monitoring, CI/CD (GitHub Actions), AWS, GCP (BigQuery), Tableau